

Blinkit Data Analysis with PowerBI

This capstone final project focuses on comprehensive data analysis of Blinkit, an online grocery platform, utilizing Power BI to extract meaningful insights and visualize data patterns.

By
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BLINKIT DATA ANALYSIS -POWER BI

Customer Behavior and Segmentation

Key Insights:

- Total Customers: 2,500 across 4 segments — Regular (639), Premium (633), New (628), and Inactive (600).
- Top Customer: Rayaana Krishna with the highest activity and value.
- Average Order Value: ₹2.20K, highest among New (₹2,287.93) and Premium (₹2,154.05) customers.
- Total Revenue: ₹11.01M, mostly driven by Premium and Regular segments.
- Customer Distribution by Pincode & Area: Dense activity from specific pincodes; cities like Jaina, Bathinda, and Deogar are top in customer counts.
- Monthly Trend: Steady decline in new customer registrations from March 2023 to April 2024 — customer acquisition needs improvement.



Rayaan Krishna

Top Customer

2.20K

Avg_order_value

5000

Total_orders

2.00

Avg Orders per Customer

11.01M

Total_Revenue

2500

Total Customers

customer_segment

All

Registration_Month

All

area

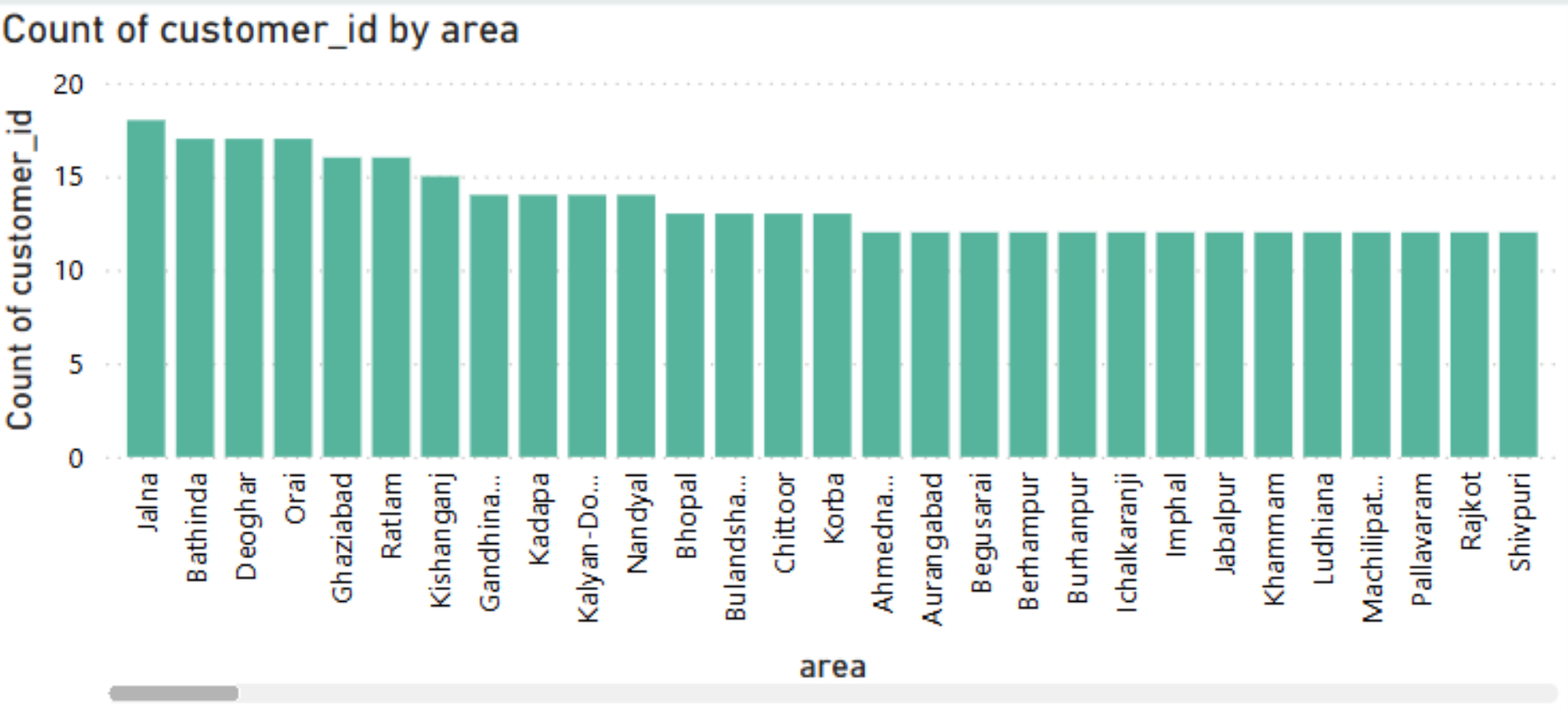
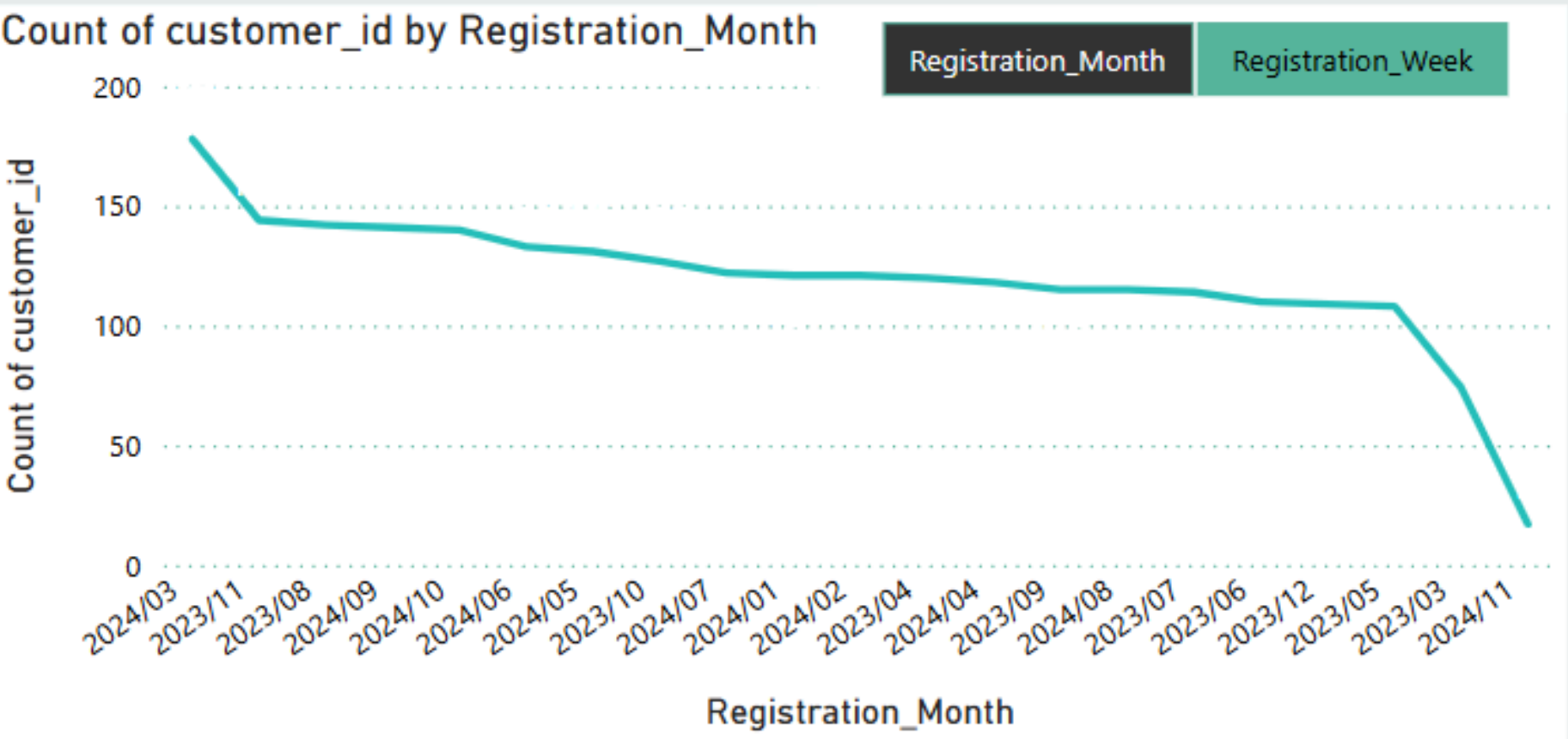
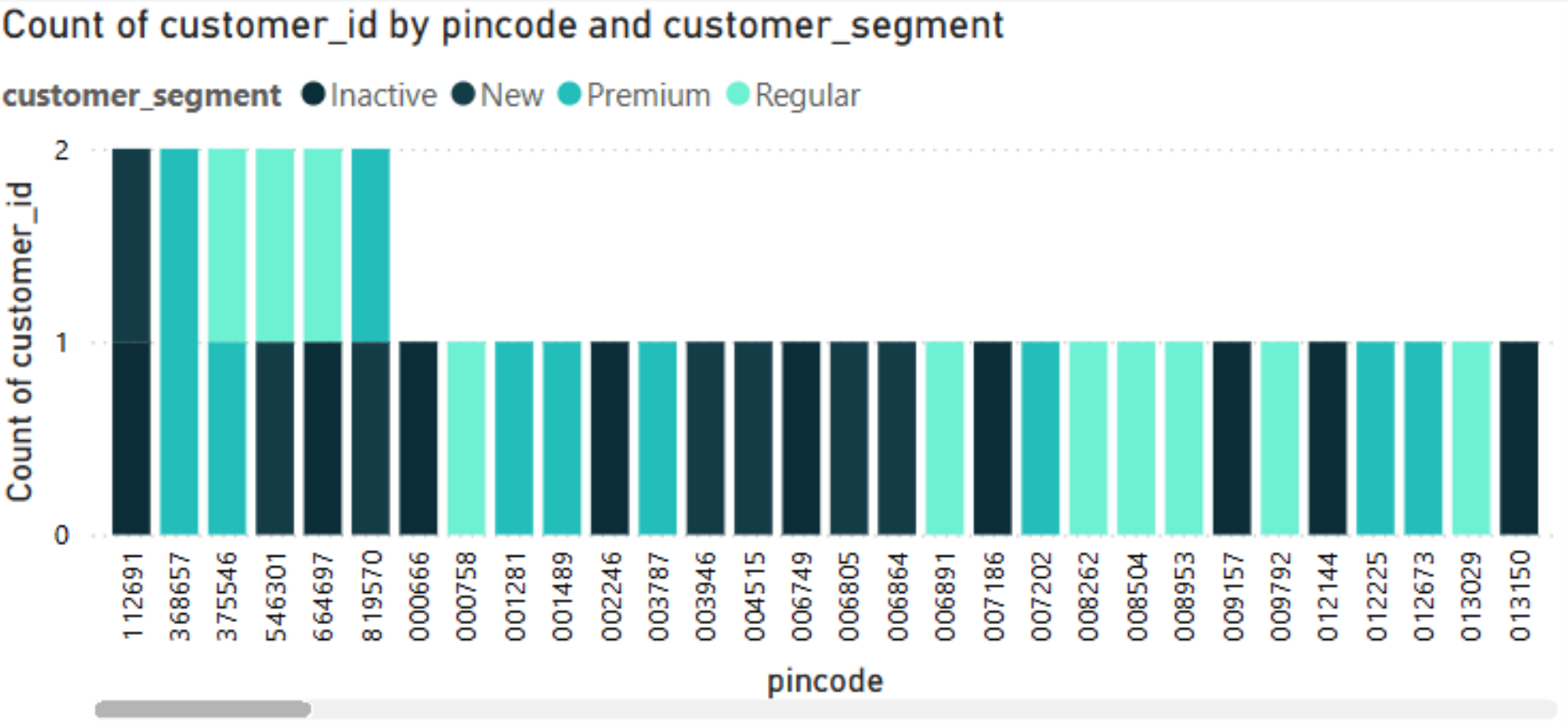
All

Registration_Week

All

customer_segment	Total Customers
Inactive	600
New	628
Premium	633
Regular	639
Total	2500

customer_segment	Avg_order_value	Total_Revenue
Regular	2,189.51	2,890,148.66
Premium	2,154.05	2,731,329.71
New	2,287.93	2,795,854.48
Inactive	2,178.13	2,591,975.65
Total	2,201.86	11,009,308.50



Order and Delivery Performance

Key Insights:

- Total Orders: 5,000 with 69.4% delivered on time, and 30.6% delayed (1,530 deliveries).
- Avg Delivery Time: 4.44 minutes. Some delivery partners have excessive times (up to 26 mins).
- Reason for Delay: 100% of delays are due to traffic.
- Delivery Partner Performance: Huge variation in partner performance — optimization and training required.
- Slightly/Significantly Delayed Orders: 1,000+ orders face minor delays; around 500 are significantly delayed — indicating systemic bottlenecks.

5000

Total Deliveries

1530

Delayed Deliveries

3470

On_time_Deliveries

4.44

Avg_Delivery_Time

30.60%

Delayed_Delivery_rate

69.40%

On_Time_Delivery_Rate

delivery_status



All



delivery_partner_id



All



payment_method



All



order_id



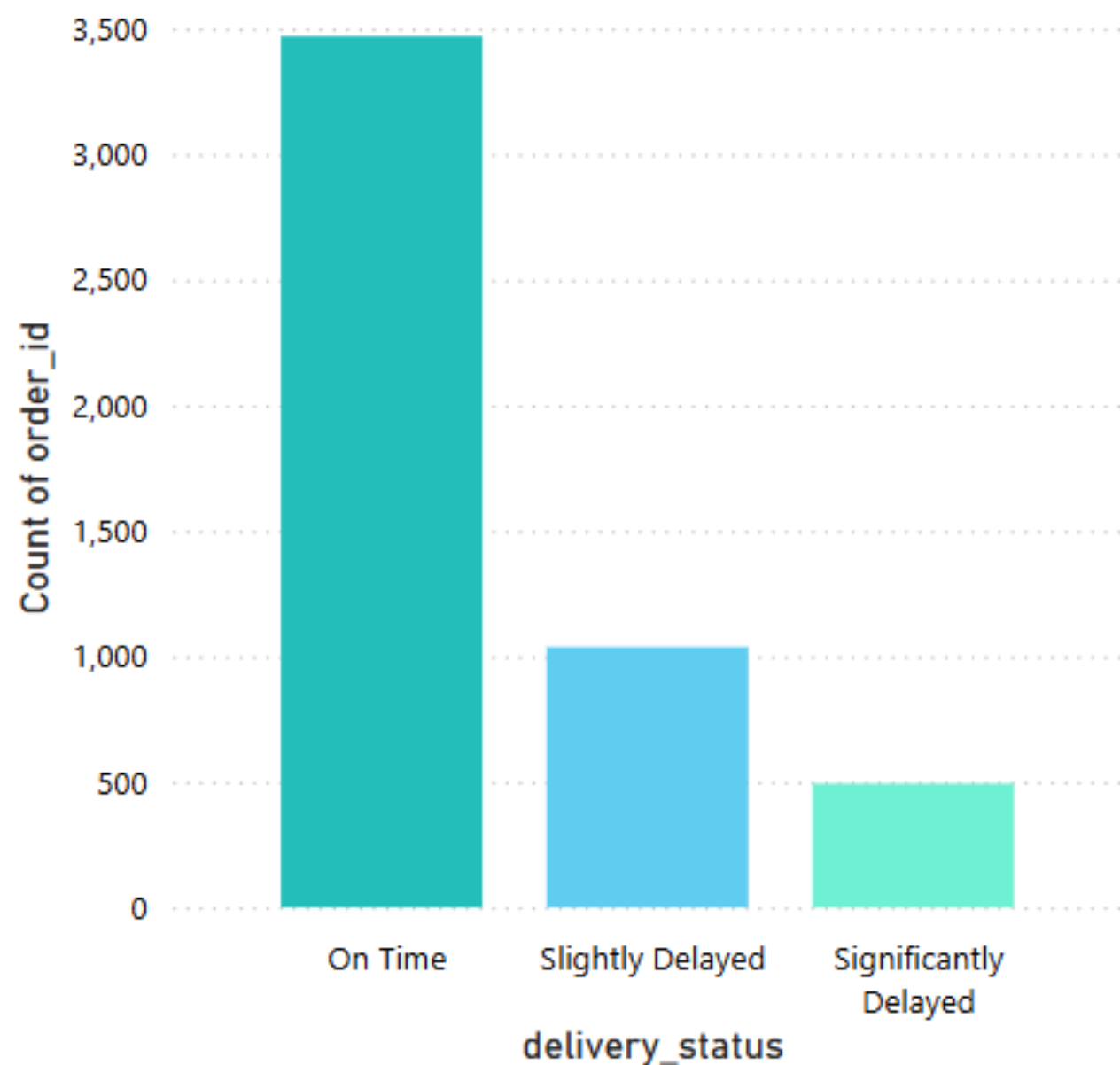
All



Avg_Delivery_Time by delivery_partner_id



Count of order_id by delivery_status



Count of order_id by reasons_if_delayed



1.53K (100%)

reasons_if_delayed

Traffic

delivery_partner_id		Avg_Delivery_Time
99968	→	12.00
99958	↓	1.00
99899	↘	4.00
99898	↘	6.00
99889	→	12.00
99870	↘	4.00
99820	↘	6.00
99775	↓	1.00
99752	→	13.00
99751	↘	6.00
99744	↘	7.00
99733	↘	2.00
99684	↑	26.00
99605	↑	23.00
99602	↓	1.00
Total		4.44

Product and Inventory Management - I

Key Insights:

- Top Products:
 - By Revenue: Baby Food
 - By Quantity: Wheat Flour
 - By Margin: Wheat Flour (40%)
- Current Stock vs Min Stock Level: Most products like Baby Food and Dish Soap are near minimum levels — possible stockout risk.
- Total Quantity Sold: Pet Treats, Toilet Cleaner, and Dish Soap lead.
- Stock Damaged: High damage in Vitamins, Cough Syrup, and Pet Treats.
- Recommendation: Strengthen supply chain checks for top-selling and fragile items.

Baby Food

Top Product by Revenue

Wheat Flour

Top Product by Quantity

27.78

Average Margin %

40

Top Product by Margin %

Wheat Flour

Top Product Name by Margin

99.51%

StockRotationRate

820

Sum of Current_Stock

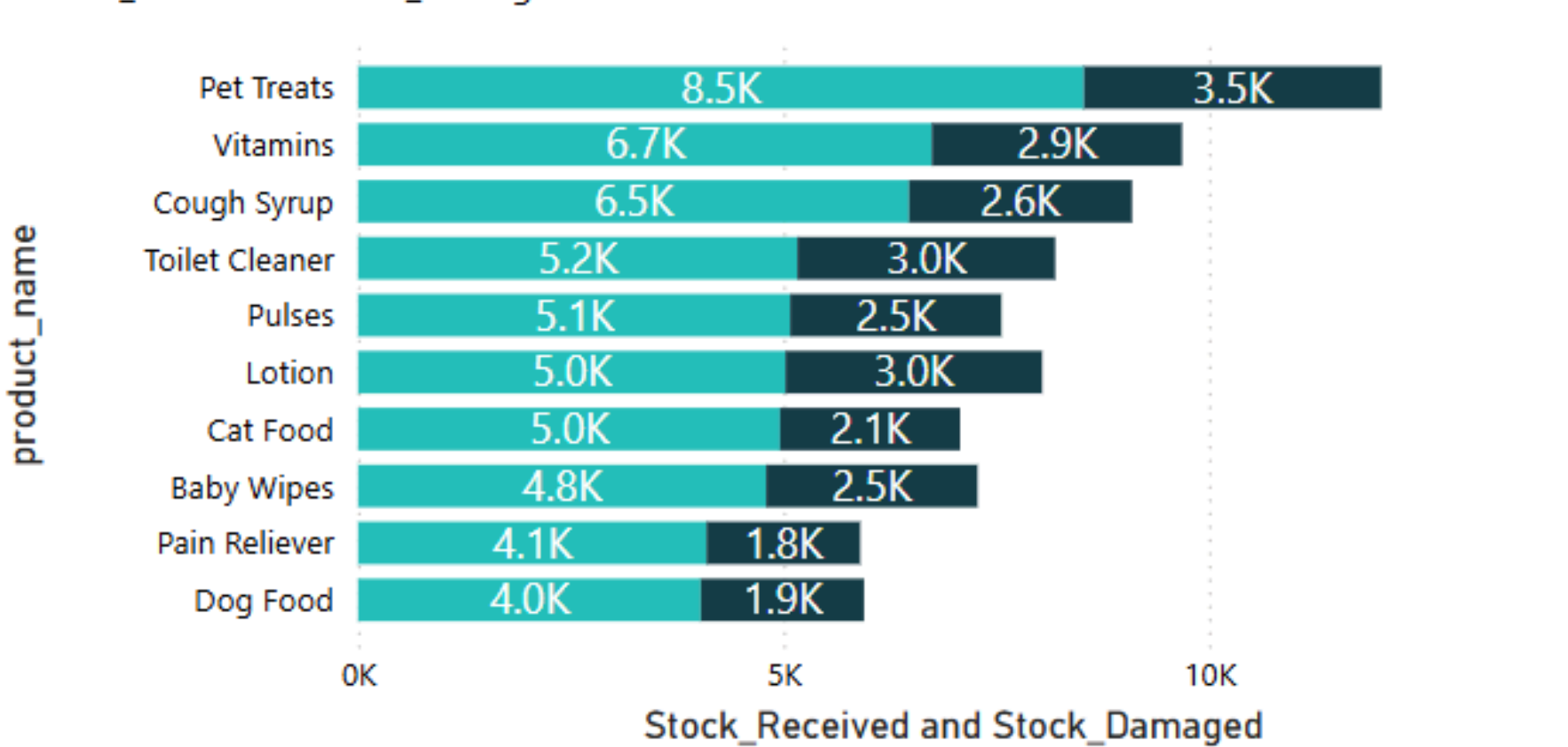
brand	product_id	First product_name	Sum of min_stock_level	Sum of Current_Stock
All	4452	Baby Wipes	27	
	6405	Baby Food	11	4
category	9436	Toothpaste	19	
All	11422	Potatoes	20	12
	14145	Spinach	14	4
	15314	Cheese	30	4
date	18035	Dish Soap	12	8
All	26060	Vitamins	25	8
	33797	Pain Reliever	12	
stock_r...	33955	Frozen Pizza	29	4
4	Total	Baby Food	5464	820

Total Quantity Sold by product_name



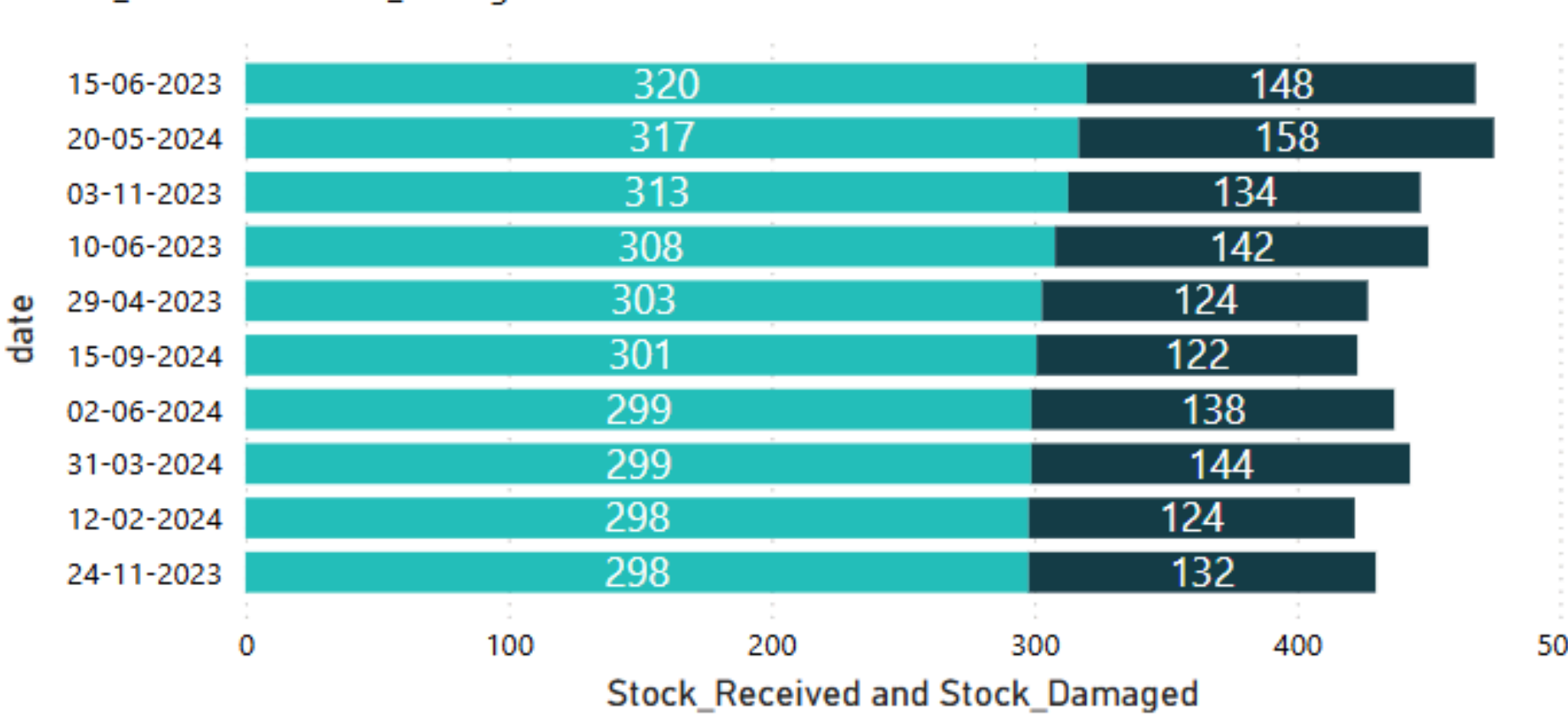
Stock_Received and Stock_Damaged by product_name

● Stock_Received ● Stock_Damaged



Stock_Received and Stock_Damaged by date

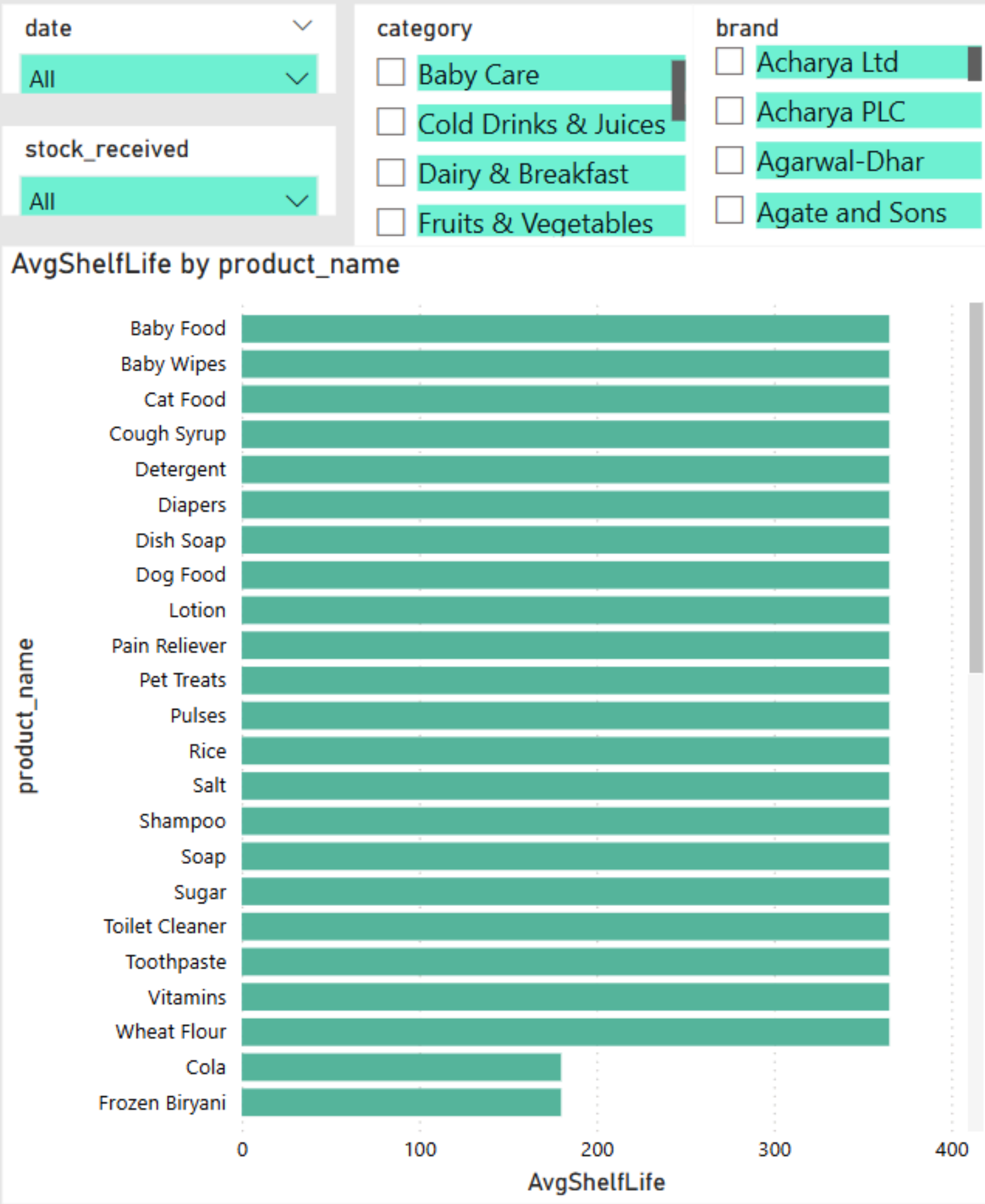
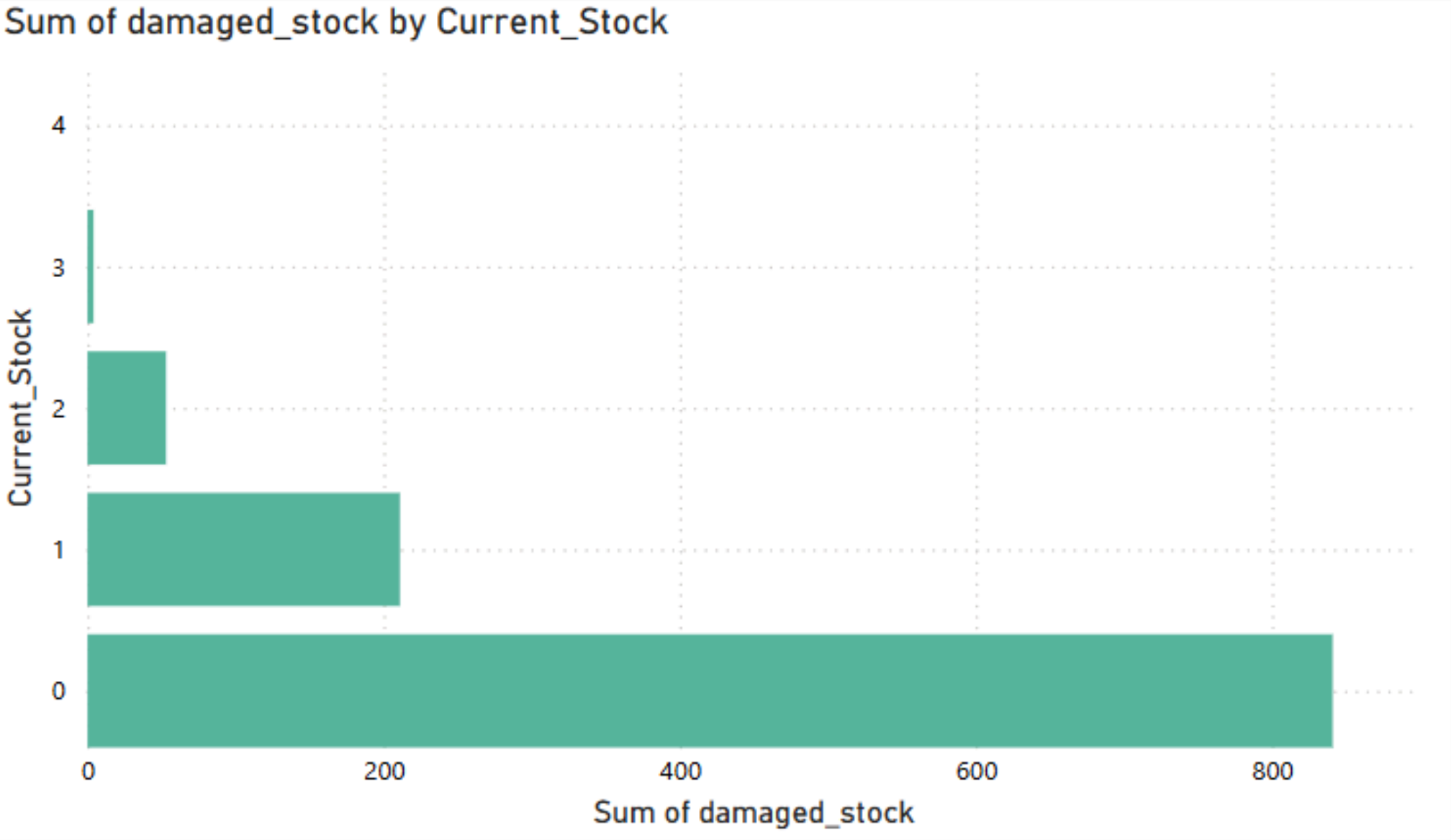
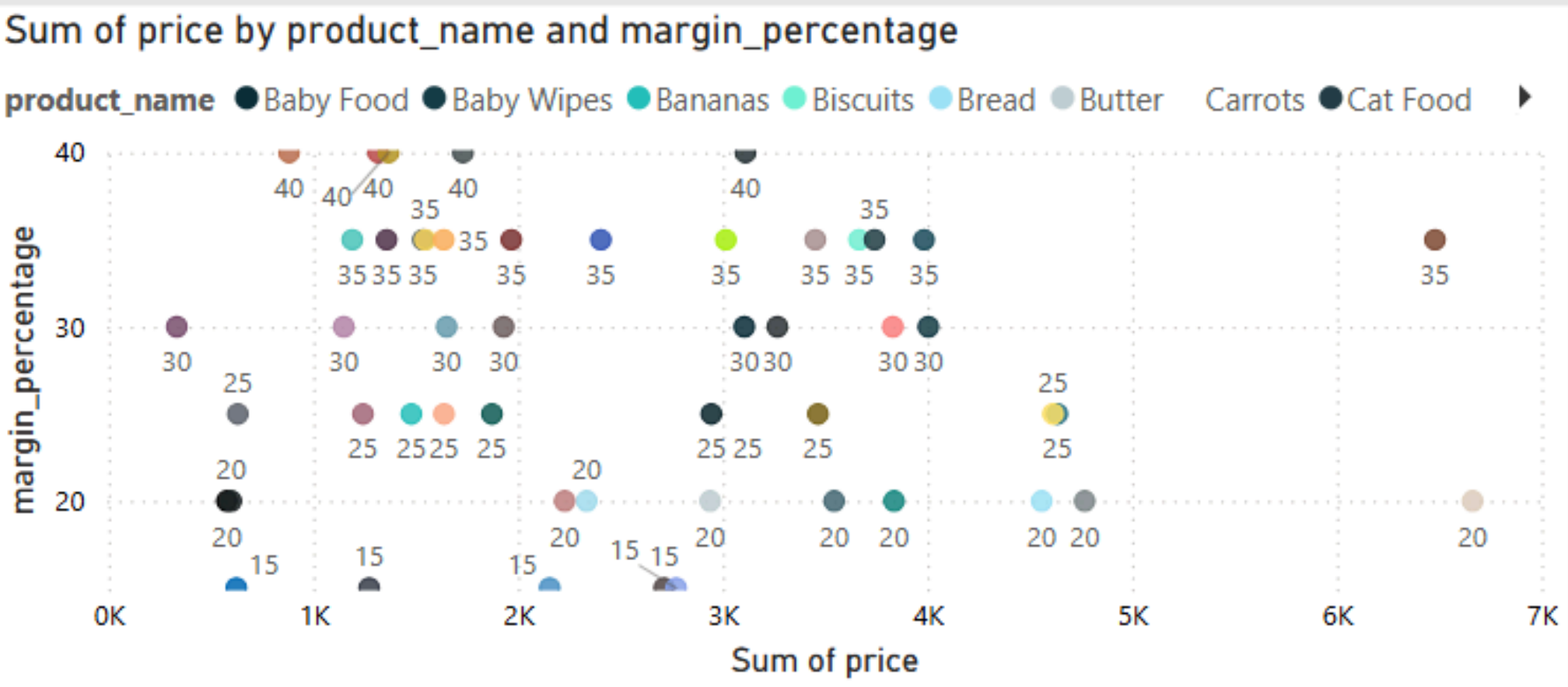
● Stock_Received ● Stock_Damaged



Product and Inventory Management - II

Key Insights:

- Price vs Margin: High-margin products like Baby Food, Bananas, and Biscuits need focused promotion.
- Current vs Damaged Stock: Large portion of stock in categories like Baby Wipes and Pain Relievers is damaged — resulting in loss.
- Shelf Life Analysis: Most products have good shelf life (300+ days), except Frozen Biryani and Cola which are lower — should be sold quickly.
- Recommendation: Improve packaging and handling to reduce damage; optimize pricing for products with high margin and long shelf life.



Marketing Campaign Performance

Key Insights:

- Revenue Generated: ₹32.19M from multiple campaigns.
- Total Conversions: 298K with only 1.01% conversion rate — large gap between impressions and actual conversions.
- Top Channels by ROAS (Return on Ad Spend): All channels average around 2.7 ROAS. App and Email perform slightly better.
- Conversion Trends:
 - New User Discount & Referral Programs perform better with New Users.
 - Push notifications and Flash Sales have lower effectiveness.
- Recommendation: Refocus campaigns toward high-converting segments (Premium & New users) and reduce budget on underperforming channels.

₹32.19M	1.01%	16.32M	14.80K	29M	298K	3M
Sum of revenue generated	Conversion Rate	Total Spend	Roas	TotalImpressions	TotalConversions	TotalClicks

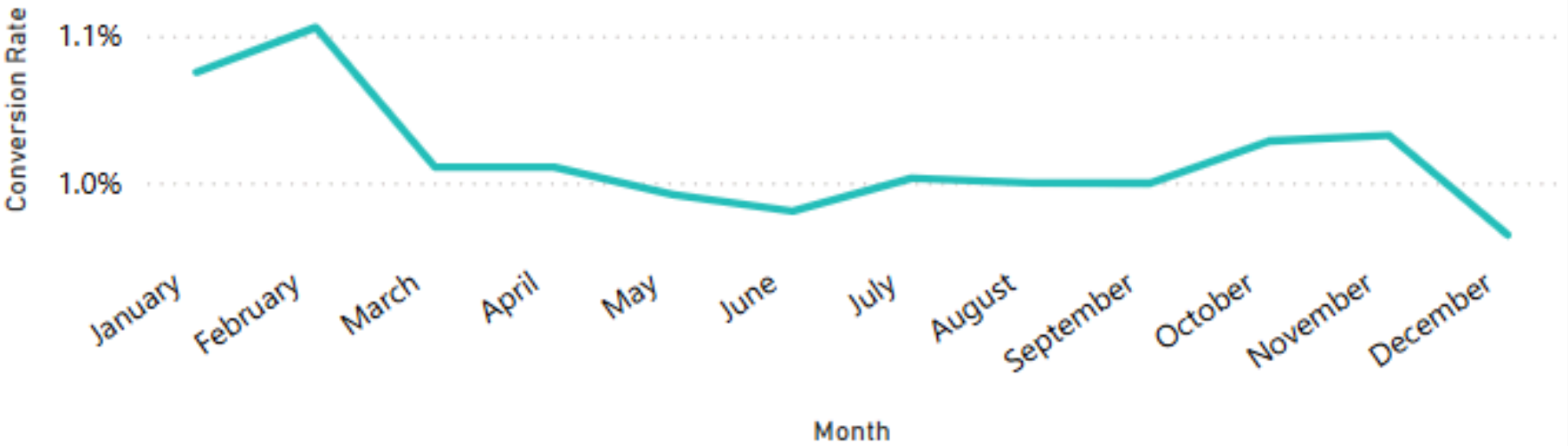
campaign_name	channel	target_audience
All	All	All

campaign_name	Sum of impressions	Sum of clicks	Sum of conversions	Sum of spend
App Push Notification	3130007	325833	3278300%	1,788,989.20
Flash Sale	3264022	332882	3321100%	1,831,687.82
New User Discount	3270347	332400	3396400%	1,833,454.81
Referral Program	3276426	329302	3349000%	1,818,025.51
Weekend Special	3200521	322668	3227100%	1,700,611.20
Total	29487610	2974145	29803800%	16,319,838.24

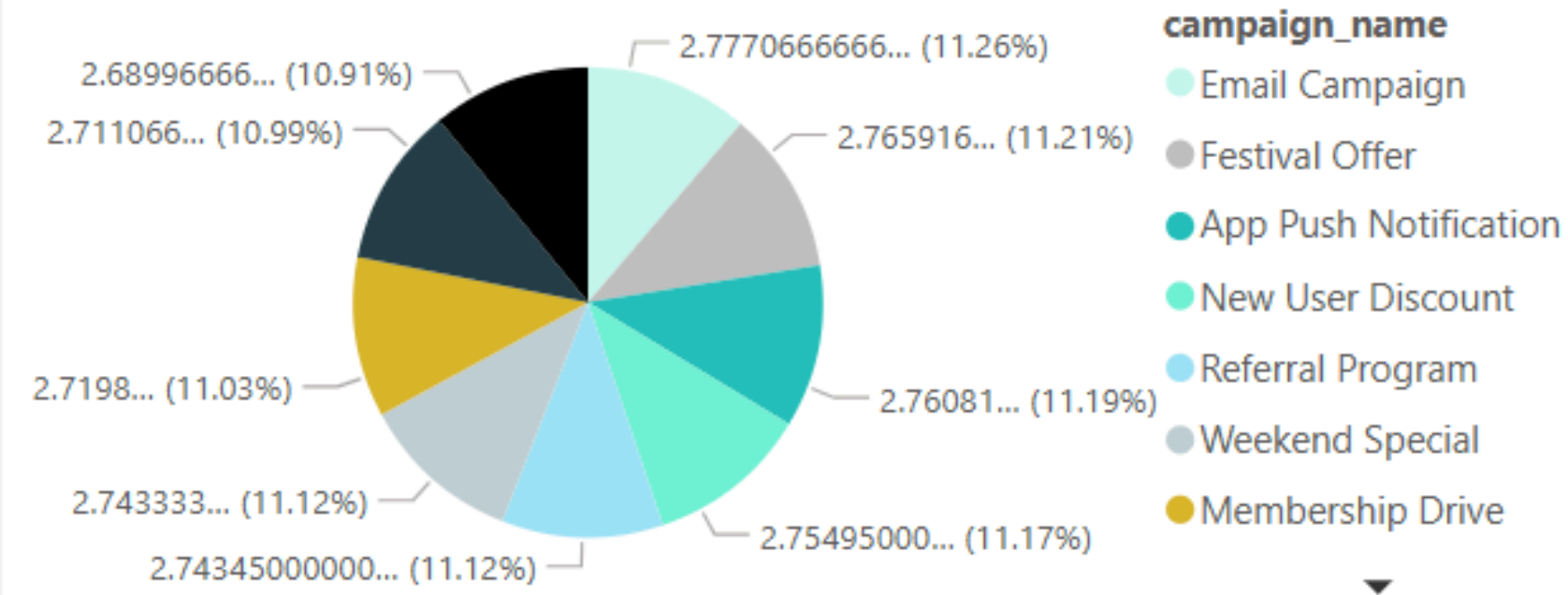
channel	Sum of impressions	Sum of clicks	Sum of conversions	Sum of spend	Average ROAS
App	7563662	734975	7519200%	4,213,378.75	2.75
SMS	7367010	744274	7323500%	3,998,607.54	2.71
Social Media	7281244	738839	7494000%	4,110,363.91	2.75
Email	7275694	756057	7467100%	3,997,488.04	2.75
Total	29487610	2974145	29803800%	16,319,838.24	2.74

target_audience	Sum of impressions	Sum of clicks	Sum of conversions	Sum of spend	Average
All	7290747	739244	7453700%	4,110,307.60	
Inactive	7277898	735304	7414200%	4,077,275.86	
New Users	7495975	745239	7575600%	4,082,313.37	
Premium	7422990	754358	7360300%	4,049,941.41	
Total	29487610	2974145	29803800%	16,319,838.24	

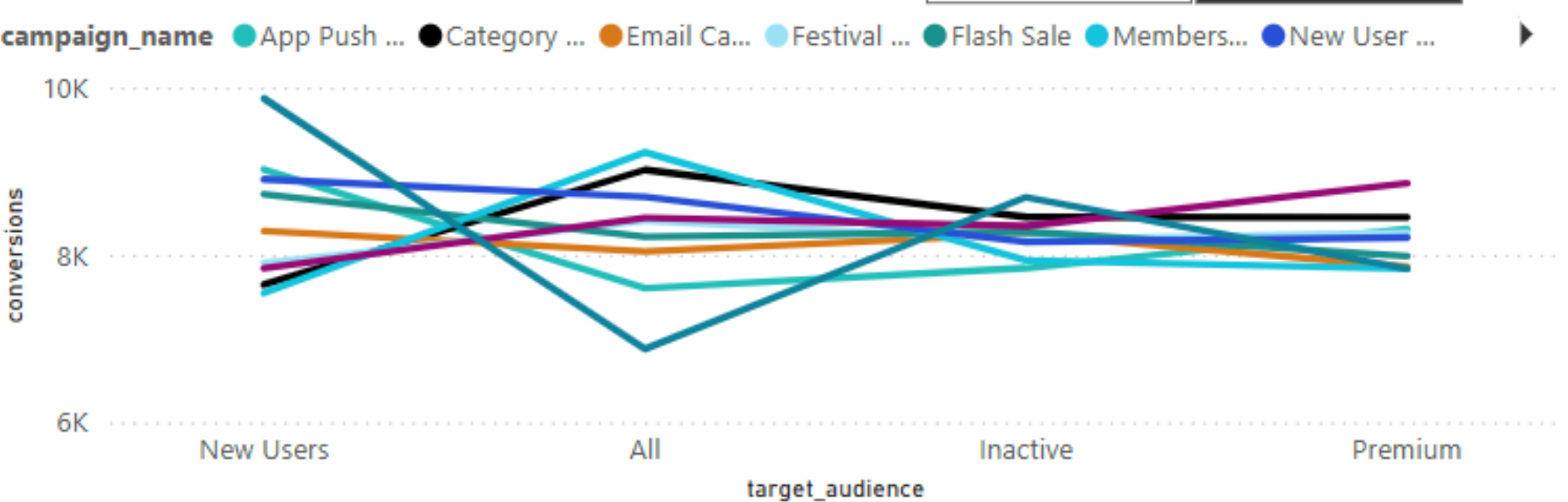
Conversion Rate by Month



Average ROAS by campaign_name



conversions by target_audience and campaign_name



Customer Feedback and Sentiment Analysis - I

Key Insights:

- Feedback Count: 5,000 responses with a 32.84% negative rate.
- Average Rating: 3.34 out of 5. Delivery and Product Quality rated lowest (3.3).
- Sentiment vs Delivery: Positive sentiments mostly tied to on-time delivery; delays sharply correlate with negative reviews.
- Monthly Sentiment Trend: May and July show peaks in feedback — possible product or service issues during these months.
- Recommendation: Prioritize improvements in delivery handling and product quality assurance.

1620

TotalPositiveFeedback

1642

TotalNegativeFeedback

1738

Neutral Feedback

5000

FeedbackCount

3.34

AverageRating

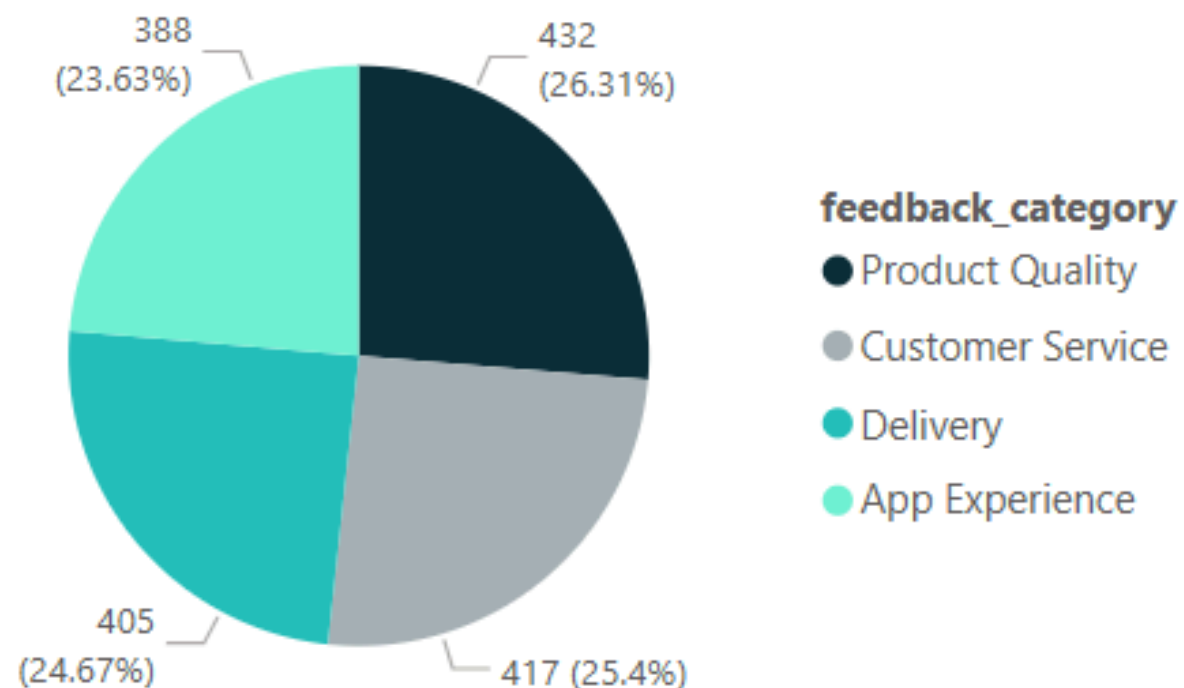
32.84

Negative Feedback %

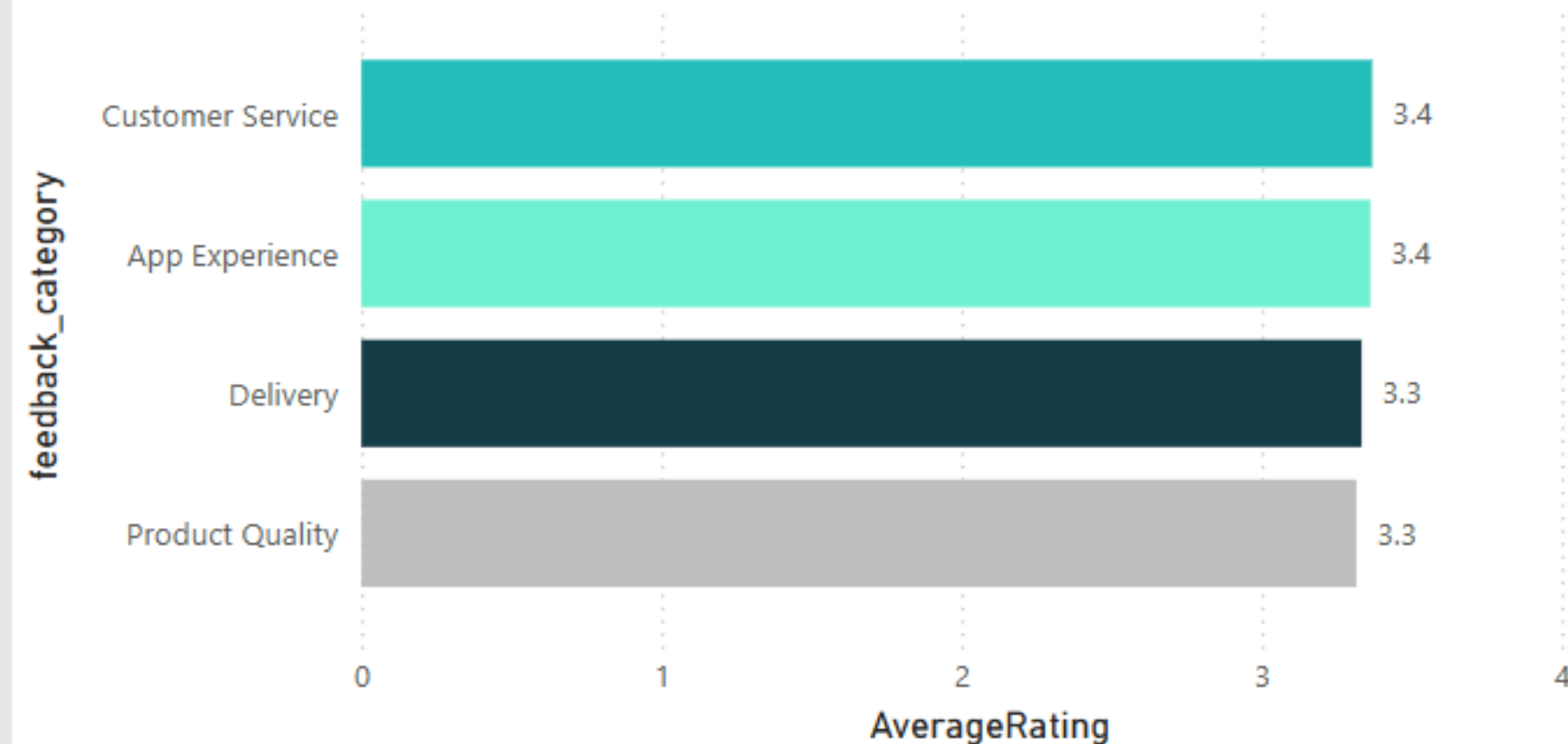
2.30

Feedback per Customer

TotalNegativeFeedback by feedback_category

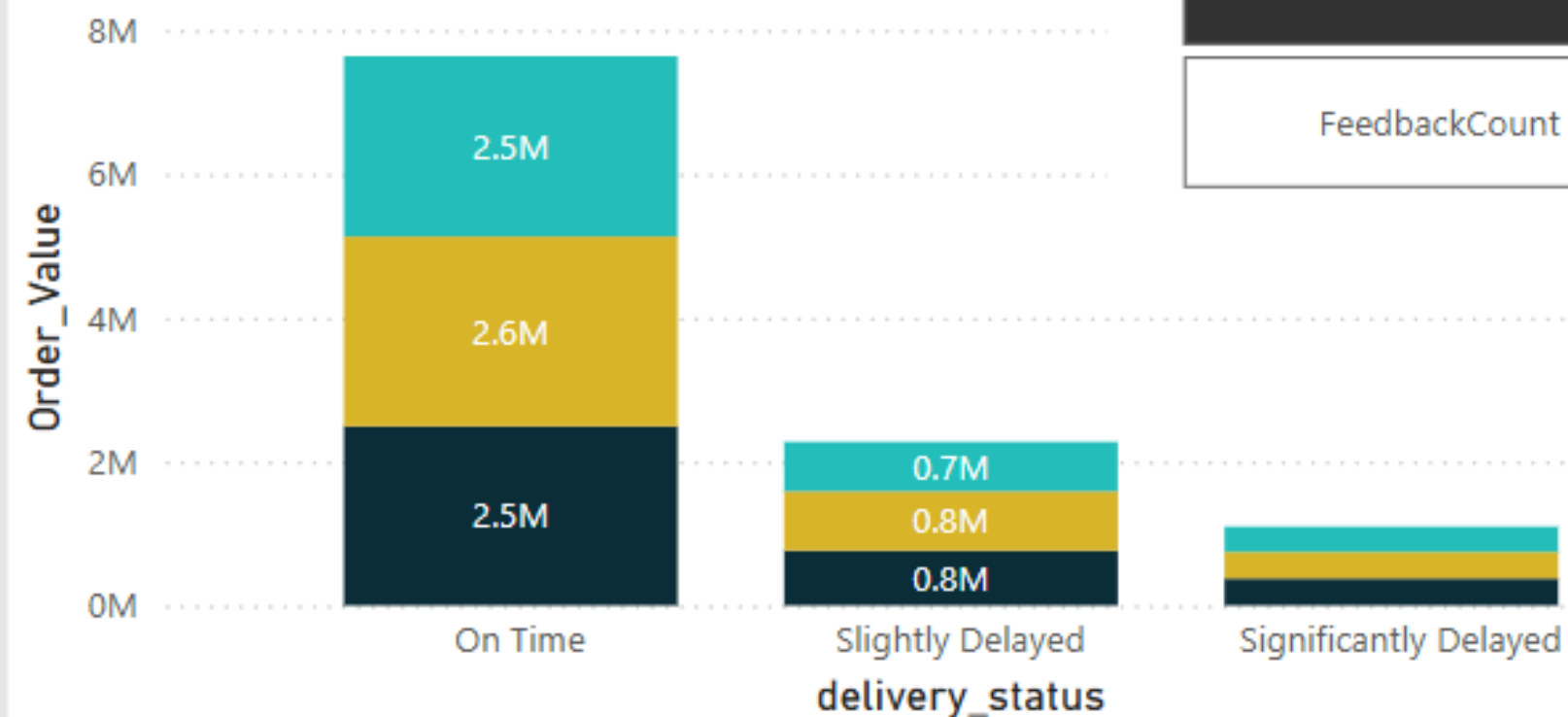


AverageRating by feedback_category



Order_Value by delivery_status and sentiment

sentiment ● Negative ● Neutral ● Positive



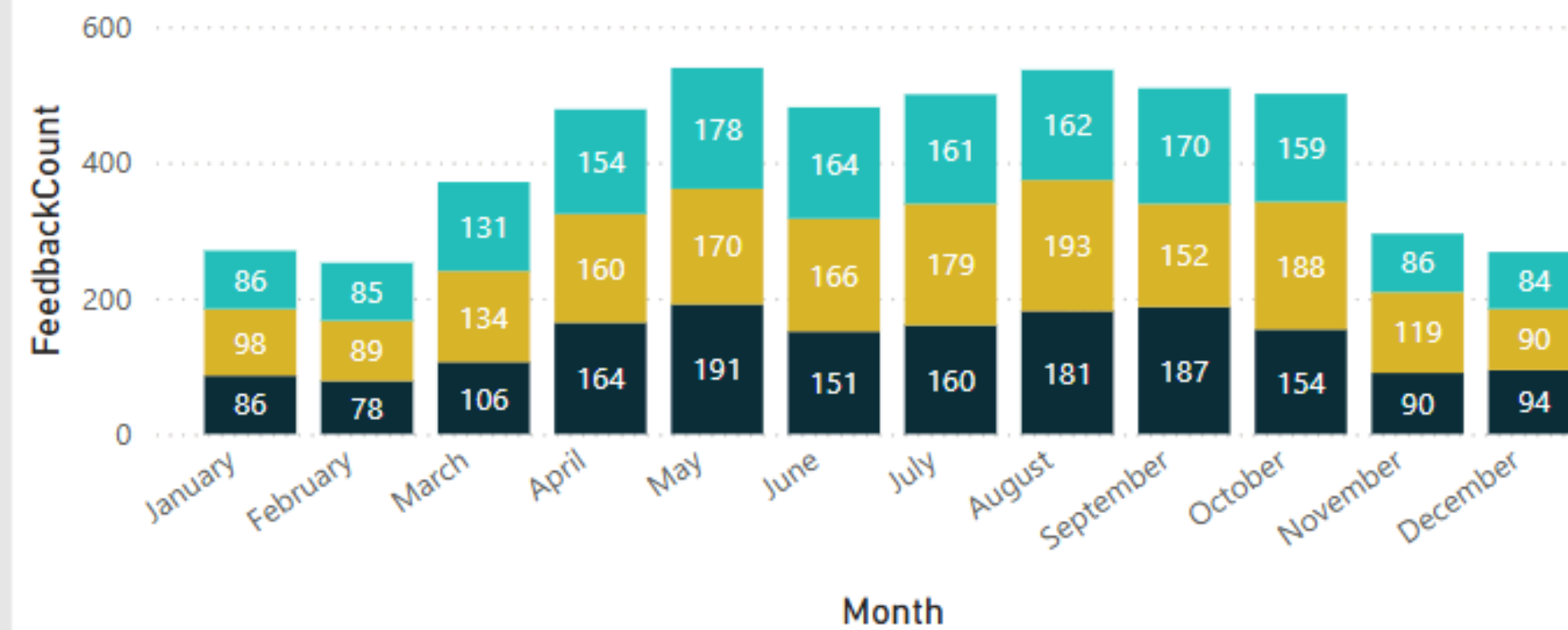
Parameter 6

Order_Value

FeedbackCount

FeedbackCount by Month and sentiment

sentiment ● Negative ● Neutral ● Positive



Customer Feedback and Sentiment Analysis - II

Key Insights:

- Sentiment Split: 34.5% Positive, 33.2% Neutral, 32.3% Negative — almost balanced.
- Frequent Complaints:
 - “Items were missing”
 - “Product was damaged during delivery”
 - “Packaging was poor”
- Feedback Word Cloud: Confirms recurring negative experiences related to logistics and customer service.
- Recommendation: Address repeat complaint patterns through customer support training and warehouse checks.

sentiment

Neutral

Positive

feedback_category

Delivery

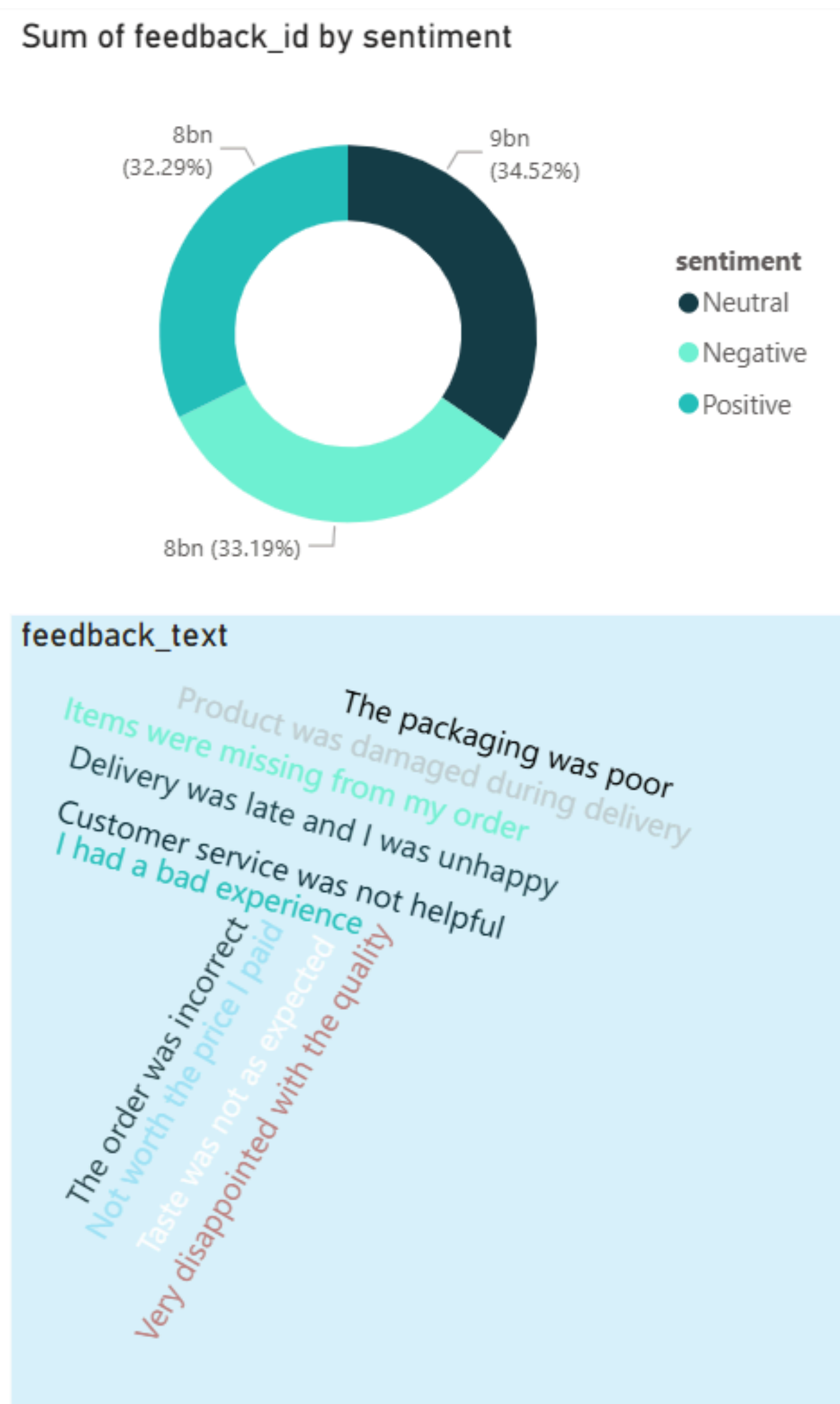
Product Quality

feedback_date

4/5/2023

4/6/2023

customer_name	order_id	Year	Month	Day	sentiment	feedback_text
Aarna Naik	3163611427	2024	January	17	Negative	Items were missing from my order.
Aayush Sengupta	957188879	2024	January	12	Negative	Not worth the price I paid.
Anika Raja	299859815	2024	January	16	Negative	Product was damaged during delivery.
Anya Panchal	4590662628	2024	January	10	Negative	Not worth the price I paid.
Arjun Deep	6392348668	2024	January	29	Negative	Product was damaged during delivery.
Aryan Ghose	2065203676	2024	January	15	Negative	I had a bad experience.
Aryan Ravel	8290493339	2024	January	29	Negative	I had a bad experience.
Bakhshi Sodhi	349410429	2024	January	29	Negative	The packaging was poor.
Bhavna Ramaswamy	189431942	2024	January	23	Negative	Product was damaged during delivery.
Bhavya Dutta	6703651257	2024	January	17	Negative	Delivery was late and I was unhappy.
Bhavya Tiwari	9103834515	2024	January	3	Negative	The packaging was poor.
Brinda Datta	8545716586	2024	January	12	Negative	Delivery was late and I was unhappy.
Chakrika Sachar	6129180304	2024	January	17	Negative	Customer service was not helpful.
Chandran Sur	9416522701	2024	January	22	Negative	Items were missing from my order.
Charan Choudhary	4217659083	2024	January	18	Negative	Product was damaged during delivery.
Charvi Bajaj	7891517071	2024	January	31	Negative	I had a bad experience.
Damyanti Nadkarni	9057909300	2024	January	17	Negative	Items were missing from my order.
Darika Raj	7415158732	2024	January	5	Negative	I had a bad experience.
David Thaker	7902025520	2024	January	2	Negative	Very disappointed with the quality.
Falan Dhillon	9195687538	2024	January	6	Negative	The packaging was poor.
Farhan Khare	3559865135	2024	January	7	Negative	Not worth the price I paid.
Harish Mahajan	1677502866	2024	January	18	Negative	Product was damaged during delivery.
Harita Gulati	7834171020	2024	January	4	Negative	Items were missing from my order.
Hema Luthra	1708282193	2024	January	25	Negative	Product was damaged during delivery.



Financial and Operational KPIs

Key Insights:

- Total Revenue: ₹11.01M
- Order Volume: 5,000 orders
- Avg Orders per Customer: 2.0 — indicates moderate repeat behavior.
- Top Category by Margin: Instant & Frozen Food
- Damaged Stock Cost: ₹0.53M (~4.6% of revenue) — significant loss.
- Monthly Revenue Trend: Peak revenue observed in May–September. Revenue dips sharply after October.
- Payment Method Split: Even distribution across UPI, Wallet, Card, and Cash — no dominant mode.
- Recommendation: Encourage customer retention with loyalty programs and reduce damaged stock losses.

Instant & Frozen Food

TopCategoryByMargin

7445

Margin_Percentage

40

MaxMargin

11.01M

Order_Value

5000

Total_orders

0.69

Repeat_Customer_Rate

2.00

Avg Orders per ...

sentiment

All

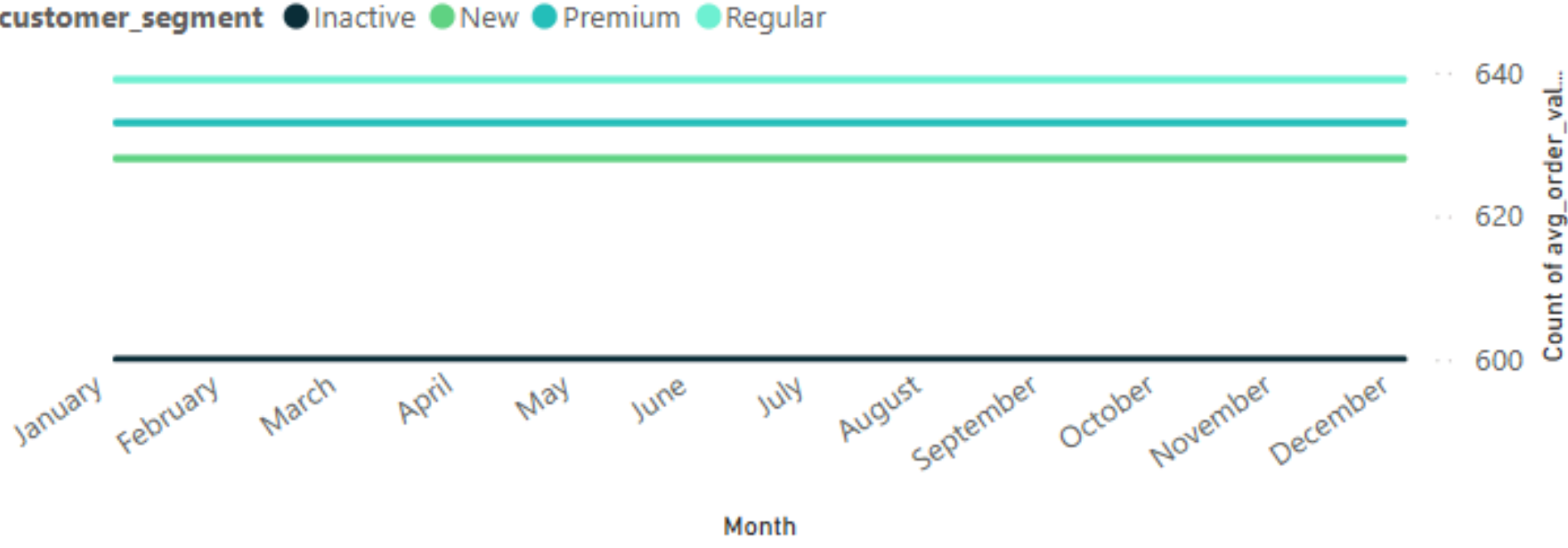
customer_segment

All

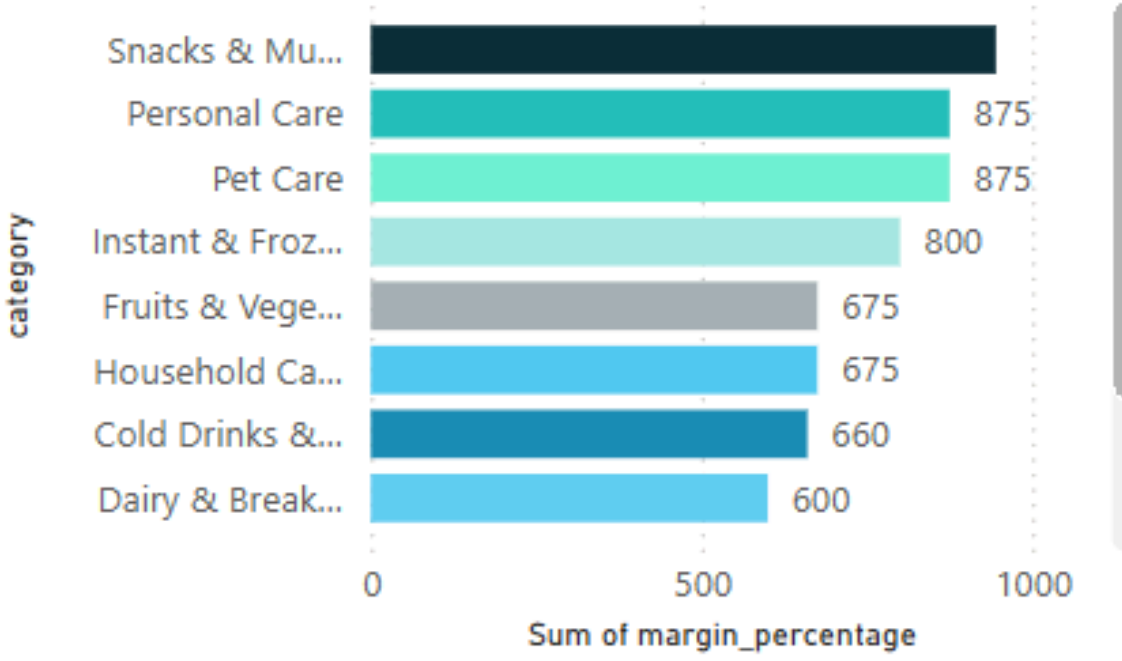
Month Name

All

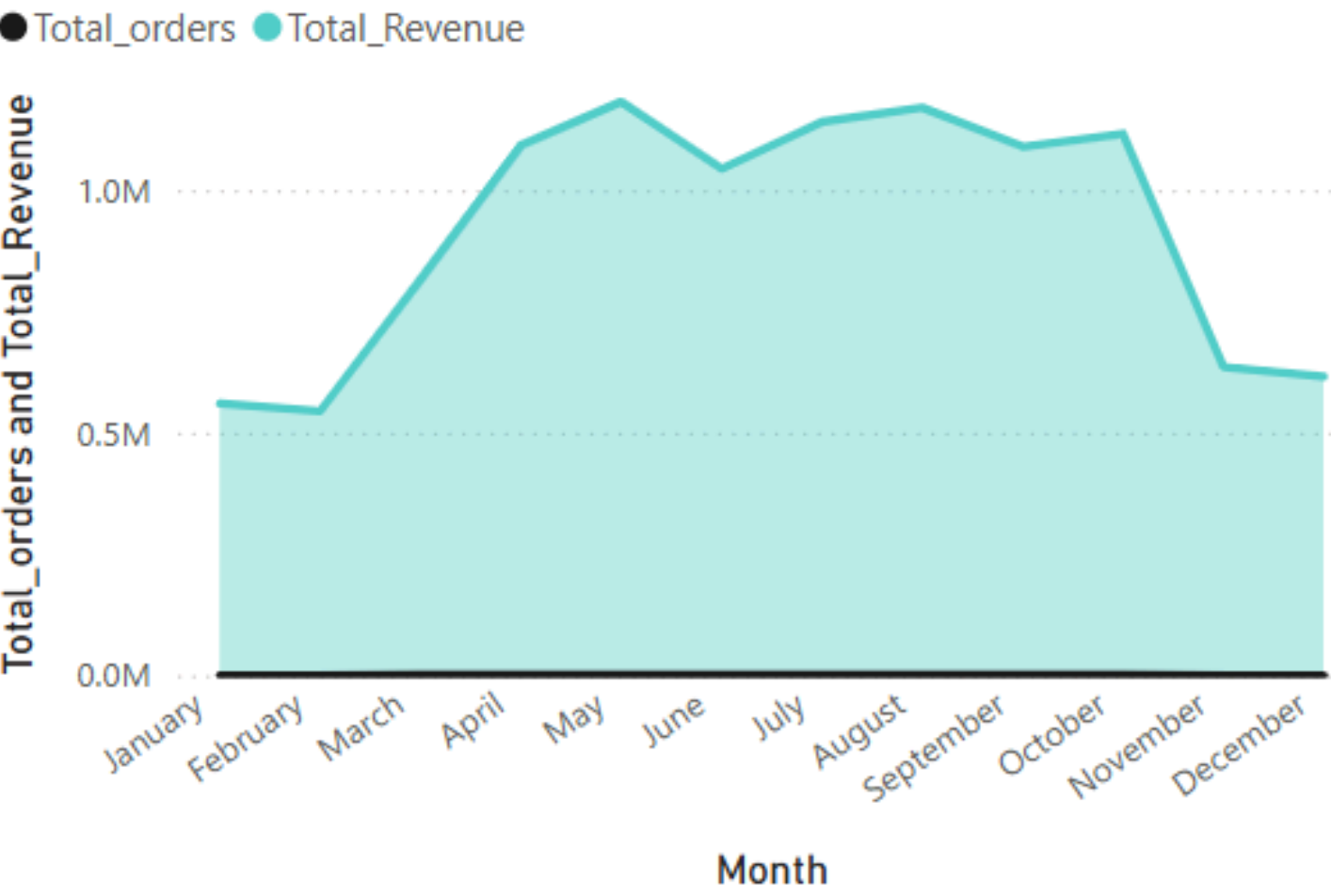
Count of avg_order_value by Month and customer_segment



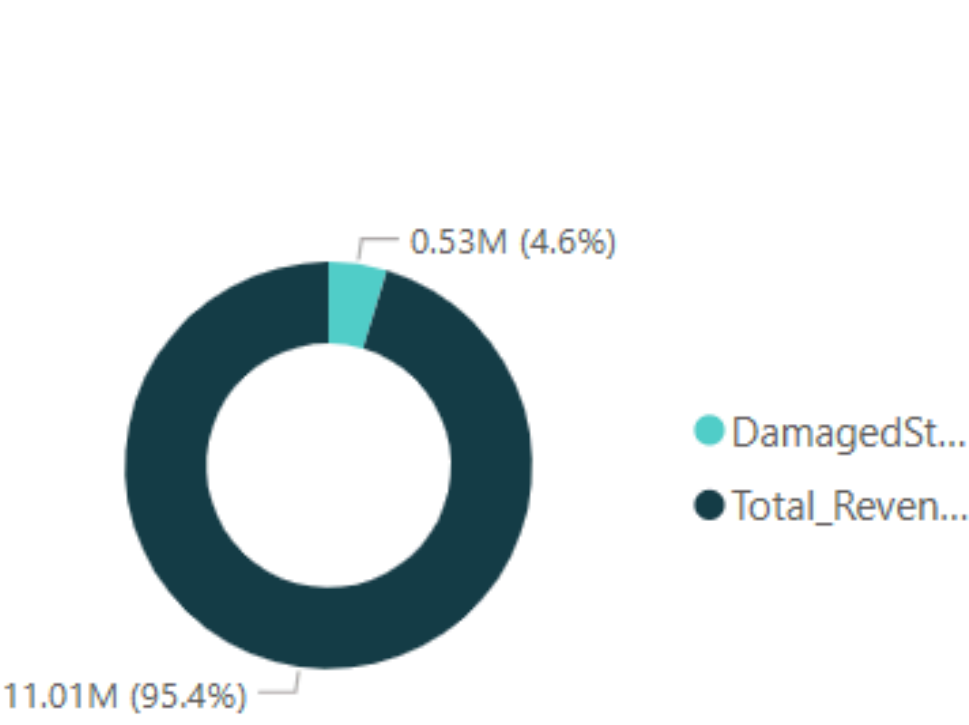
Sum of margin_percentage by category



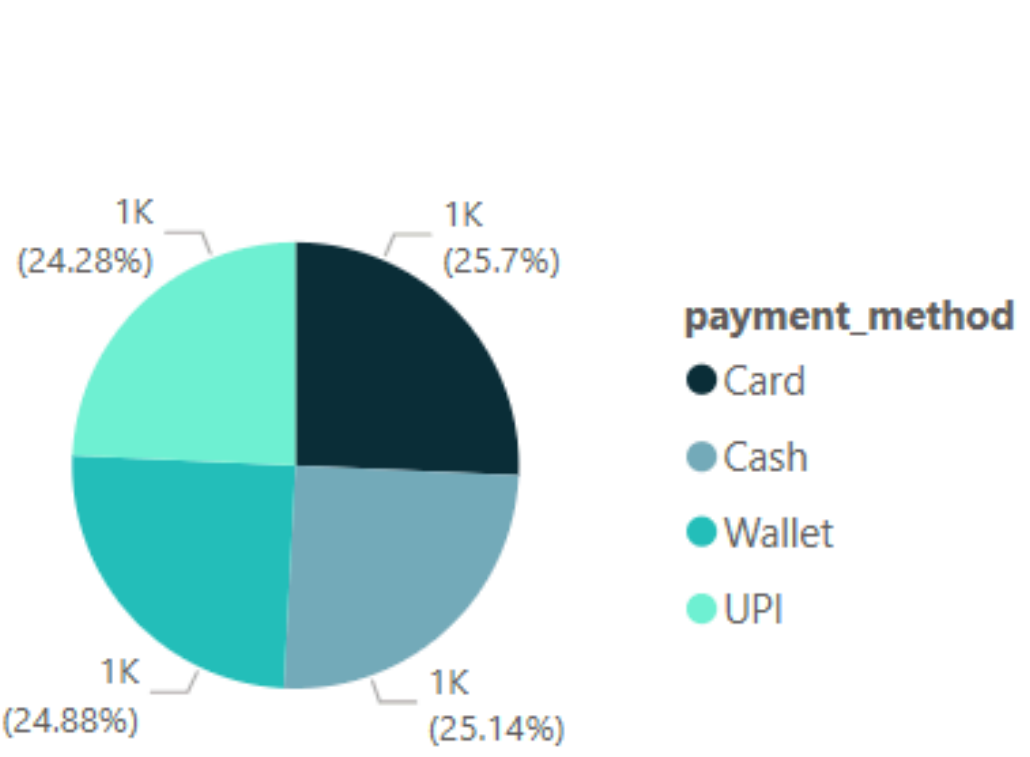
Total_orders and Total_Revenue by Month



DamagedStockCost and Total_Revenue



Total_orders by payment_method



Conclusion

This analysis gave us a clear picture of how Blinkit is performing across different areas – from customer behavior and deliveries to marketing and inventory. While the business is generating solid revenue and has a decent customer base, there are clear opportunities for improvement. Delayed deliveries, product damages, and low marketing conversions are costing time and money. At the same time, high-margin products and loyal customers show a lot of potential. If Blinkit focuses on fixing delivery issues, reducing stock damage, and targeting the right users in campaigns, it can grow faster and serve customers better. With these insights, the company is in a strong position to make smarter decisions moving forward.